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Examining the Influence of AI-Supporting HR Practices Towards Recruitment Efficiency with the Moderating Effect of Anthropomorphism

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Abstract: Technological developments are compelling organizations to upgrade their HR practices by adopting AI-driven applications. Yet, HR professionals are hesitant to adopt AI-driven technology in the recruitment process. Addressing this topic, the current study developed an amalgamated research framework for investigating factors relevant to AI, such as perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience, which was applied to investigate employees' intention to adopt AI-driven recruitment. For our data collection, survey questionnaires were distributed among HR professionals, which garnered 336 respondents. The empirical findings revealed that perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience explained a large portion (89.7%) of the variance R^2 in employees' intention to adopt AI-driven recruitment practices. The effect size f^2 analysis, then demonstrated that perceived interactivity was the most influential factor in employees' intention to adopt AI-driven recruitment. Overall, this study indicates that perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience are the core factors enhancing employees' intention to adopt AI-enabled recruitment and should hence be the focuses of policymakers' attention. Furthermore, this study uniquely unveils a new research framework that may be applied to improve the recruitment process in organizations by using artificial intelligence, which may empower HR professionals to hire the right staff efficiently and cost-effectively. Similarly, this study is in line with United Nations sustainable development goals and contributes to decent work, industry innovation, and sustainable economic growth by using artificial intelligence human resource practices.

Keywords: perceived interactivity; perceived intelligence; personalization; real-time experience; adoption of AI-enabled recruitment; recruitment efficiency; anthropomorphism; sustainable development goals

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Academic Editor: Grigorios L. Kyriakopoulos

Received: 20 January 2025
Revised: 24 February 2025
Accepted: 14 March 2025
Published: 17 March 2025

Citation: Dalain, A.F.; Yamin, M.A. Examining the Influence of AI-Supporting HR Practices Towards Recruitment Efficiency with the Moderating Effect of Anthropomorphism. *Sustainability* **2025**, *17*, 2658. https://doi.org/10.3390/su17062658

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1. Introduction

Organizations are continuously striving to recruit the right people to achieve their strategic goals, but this is a challenging process that requires ample time, resources, and financial assistance. Artificial-intelligence-driven recruitment is a recent development, wherein AI applications assist HR professionals in identifying, screening, and evaluating the most appropriate candidates for the job [1]. These AI applications are considered valuable, as rigorous screening using AI automation [2,3] is integrated with neuroscience tools in ACS style. Ref. [1] offers an unbiased approach to finding the right candidate.

In the process, AI-driven recruitment applications interact with applicants by using the latest AI chatbot tools, such as language models for dialog applications (LaMDAs), and engage potential candidates in the recruitment drive [1,4]. An AI-driven recruitment process boosts organizational efficiency by monitoring, documenting, and recording the core HR functions involved [2]. Accordingly, AI has been applied by various recruitment portals, such as LinkedIn, Glassdoor, and Indeed; however, the use of AI-driven recruitment among organizations remains underdeveloped. In this regard, ref. [2] asserted that HR professionals lack a comprehensive understanding of AI-enabled recruitment applications. In this context, it is crucial to understand which factors encourage HR practitioners to adopt AI applications in order to achieve optimal performance in their recruitment process.

The fast-paced emergence of artificial intelligence has transformed workplaces around the globe. Amid this digital surge, organizations are beginning to apply artificial intelligence in their decision-making processes [3]. As part of this trend, organizations are compelled to upgrade their HR practices with AI-driven systems. Nevertheless, HR professionals are hesitant to adopt AI technology in the recruitment process [1,2]. To address this topic, the current study developed an integrated research model for investigating employees' intentions to adopt artificial-intelligence-driven recruitment systems. Previously, studies have discussed AI-driven recruitment, in general, but there has been a lack of empirical analysis [4,5]. We synthesized central factors from the literature, such as interactivity, intelligence, personalization, accuracy, automation, real-time experience, and anthropomorphism, which have been identified as influencing employees' propensity to adopt AI-enabled recruitment systems. Among these, perceived interactivity raises the degree to which a task is regarded as enjoyable, attractive, and satisfying [6,7]. Perceived intelligence is conveyed through certain characteristics of AI-enabled HR practices and encourages employees to adopt AI-enabled recruitment [8,9]. Similarly, personalization features will motivate employees to adopt AI-enabled technologies [10]. In addition, factors such as perceived accuracy, automation, and real-time experience have been recognized as key in influencing the individual's intention to adopt AI-enabled HR practices [11–15].

This study investigated anthropomorphism as a moderating factor between the intention to adopt AI recruitment and recruitment efficiency. Anthropomorphism refers to the humanistic features of artificial intelligence, such as robots and chatbots that act like agents, that engage individuals and improve interactions with employees [16]. It is assumed that humanistic features of an artificial-intelligence-driven application moderate the relationship between the adoption of AI-enabled recruitment and recruitment efficiency. However, ref. [2] illustrated a deficit in knowledge about AI-enabled recruitment systems' reception among HR professionals. To address that gap, this study investigated AI-driven recruitment phenomena in three regards. First, this study examined how the perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience impact employees' intention to adopt AI-enabled recruitment. Second, we investigated whether AI-driven recruitment practices bring efficiency to recruitment processes. Third, we explored how perceived anthropomorphism strengthens the relationship between the intention to adopt artificial-intelligence-enabled recruitment and recruitment efficiency. This research makes a novel contribution with the finding that using AI in recruitment systems is not only cost-effective and time-saving but also boosts recruitment efficiency. The research model in this study comprised multiple factors, and hence we provide a comprehensive view of the impact of AI-driven HR practices on recruitment efficiency. In terms of uniqueness, this study is in line with the United Nations sustainable development goals (SDGs). For instance, the use of artificial intelligence-driven human resource practices brings efficiency to the recruitment process and promotes decent work, industrial innovation, and economic sustainability, and hence contributes to SDGs8, SDGs9 and SDGs16. The

remainder of this article is structured as follows: a literature review, research methodology, data analysis, discussion, research contribution to theory, contribution to methods and practice, conclusion, research limitations, and future research directions. In the following section, a conceptual link is established between exogenous and endogenous factors.

2. Literature Review

2.1. Perceived Interactivity, Perceived Intelligence, and Personalization

Today, organizations' operations are being disrupted by AIs fast penetration into industries, which compels HR professionals to reframe their HR policies. In this setting, AI-driven recruitment has gained practitioners' attention, as it offers an opportunity to boost employee and organizational performance. Nevertheless, it is crucial to understand how AI-driven recruitment improves the efficiency of the recruitment process. In this regard, the literature highlights factors such as perceived interactivity, perceived intelligence, and personalization as key to impacting employees' propensity to adopt AI-driven recruitment procedures. Among these, the term 'interactivity' refers to how using artificial-intelligence-enabled human resource practices may bring feelings of interpersonal attraction, enjoyment, and satisfaction. Perceived interactivity indicates that artificial intelligence is acting as a strong medium for two-way communication and enhancing individuals' control over the actions taken to achieve the task [6,7]. In this regard, AI-enabled HR practices may boost HR professionals' perceived interactivity and lead them to regard the adoption of AI-enabled recruitment processes positively [6–8]. Similarly, perceived intelligence is identified as an important factor that raises individuals' intention to adopt artificial intelligence in recruitment. Perceived intelligence in the setting of human resource practices is defined as the degree to which AI-based HR practices are considered useful, efficient, goal-oriented, and autonomous in producing natural language and assisting employees in making the right decisions. It is inferred that a high perceived intelligence encourages employees to adopt AI-driven recruitment procedures [8,9]. Moreover, personalization empowers HR professionals to respond to users based on their use preferences and history and, in doing so, improves the efficiency of the recruitment system [8,16]. Prior studies have established that personalization features in AI applications positively influence employees' attitudes toward adopting AI-driven technologies [10]. Therefore, the following hypotheses were conceptualized:

H1. *Perceived interactivity is positively associated with the adoption of AI-enabled recruitment.*

H2. *Perceived intelligence is positively associated with the adoption of AI-enabled recruitment.*

H3. *Personalization is positively associated with the adoption of AI-enabled recruitment.*

2.2. Accuracy, Automation, and Real-Time Experience

The proliferation of AI-driven technology has brought accuracy, automation, and real-time experiences to the recruitment process, which are key in motivating employees to adopt AI-driven HR practices [11,12,14,15]. Accuracy results when the HR team receives screened and unbiased data, which enhances the quality of HR procedures and brings efficiency to the recruitment process. This accuracy may be realized from application screening to hiring, with human errors reduced throughout for optimal results [14]. Therefore, accuracy is a key factor that boosts HR professionals' confidence in adopting AI-driven recruitment procedures. As another, prior studies have revealed that automation in recruitment processes is time-saving, cost-effective, bias-free, and reduces HR professionals' workload [12,14]. Here, certain recruitment-related work, such as job posting,

candidate sourcing, applicant screening, profile verification, and interview scheduling, may be conducted through AI-enabled automation [12,14]. Beyond that, a related factor realized by AI-driven technology is the potential for quick responses to users and hence a real-time experience during the recruitment process. Likewise, artificial-intelligence-driven HR recruitment procedures may provide real-time feedback and, in doing so, improve recruitment efficiency [11,15], encouraging employees to adopt an AI-driven recruitment system. In sum, from the literature [11,12,14,15], this study conceptualized accuracy, automation, and real-time experience as key factors impacting employees' intention to adopt AI-driven recruitment practices. Accordingly, the following hypotheses were developed:

H4. *Accuracy is positively associated with the adoption of AI-enabled recruitment.*

H5. *Automation is positively associated with the adoption of AI-enabled recruitment.*

H6. *Real-time experience is positively associated with the adoption of AI-enabled recruitment.*

2.3. Anthropomorphism

Artificial-intelligence-driven applications are inaccessible unless organizations assist employees in understanding how AI works in practice, which may be achieved via an AI-driven assistant. This machine learning software assists employees in completing complex tasks using AI-driven applications. There are a selection of AI assistants on the market today, comprising intelligent agents with anthropomorphic characteristics, such as Microsoft's Cortana, Apple's Siri, and Amazon's Alexa. AI anthropomorphism refers to the extent to which artificial intelligence has humanistic features, that is, how robots and chatbots act like agents and engage employees in the workplace to increase their task performance [17]. Prior studies have established that anthropomorphism enriches the AI user experience through emotions and senses [17–20]. Therefore, anthropomorphism is considered an important enabler of AI-driven recruitment systems. While a recent study conducted by [8] established a strong connection between anthropomorphism and employees' intention to adopt AI-driven applications, in general, there is a deficit of research examining anthropomorphism as a moderating factor in the relationship between AI-enabled recruitment and recruitment efficiency. Addressing that gap, this study established an integrated research model (Figure 1) considering anthropomorphism as a moderator of the relationship between employees' intention to adopt AI-driven recruitment and recruitment efficiency [8,16]. Accordingly, the following hypotheses were set:

H7. *The adoption of AI-enabled recruitment is positively associated with recruitment efficiency.*

H8. *The relationship between employee intention to adopt AI-enabled recruitment and recruitment efficiency is moderated by perceived anthropomorphism.*

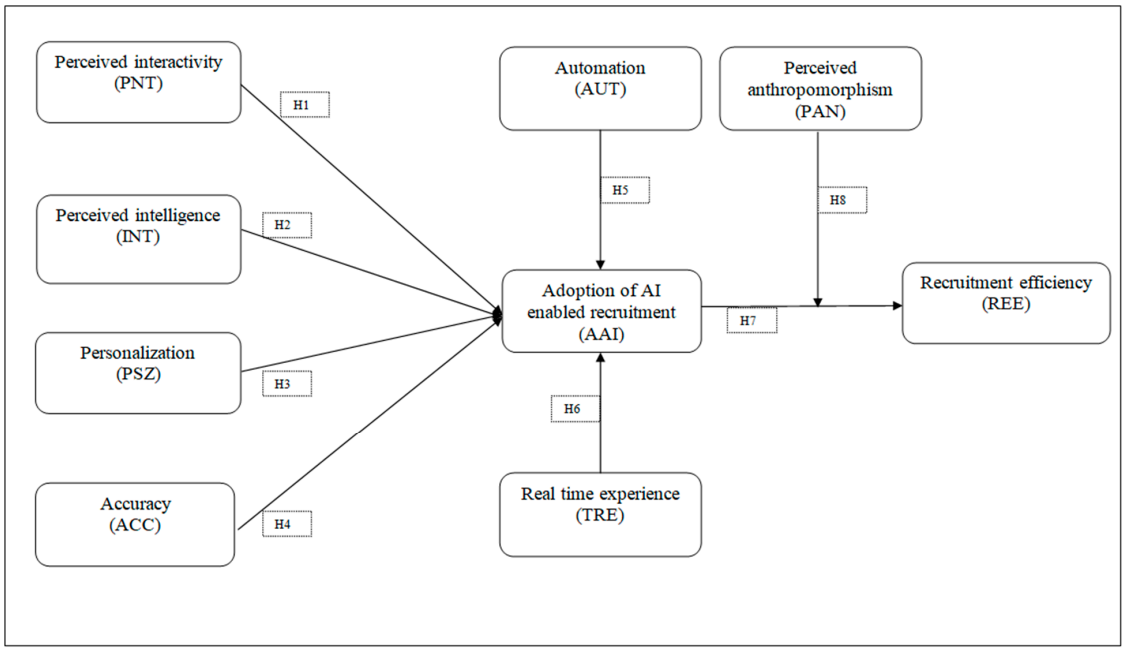


Figure 1. Research model.

3. Methodology

3.1. Methods and Instrument Development

This study followed the positivist research paradigm and applied a quantitative research method [21]. In positivist research, it is believed that knowledge exists, and therefore, the job of the researcher is to establish linkages of predictors and outcome factors with existing knowledge. Therefore, a literature review was conducted to establish causal relationships among exogenous and endogenous factors. The research model comprised multiple hypotheses, which were tested with empirical observations. Empirical data were collected through survey questionnaires comprising instruments and respondent profiles. Scale instruments were adopted from past studies and slightly amended to suit the AI recruitment setting. Scale instruments were adopted for their high loading and alpha values, with their scale items expected to offer refined scales with which to measure the concepts under study. The perceived interactivity scale was adapted from [6,8]; perceived intelligence items were adapted from [8,16]; personalization was measured with scale items adapted from [8,10]; the accuracy scale was adapted from [20,22]; automation was measured with scale items adapted from [17,20]; the real-time experience was measured with scale items adapted from [11]; the adoption of AI-enabled recruitment was measured with scale items established by [20]; scale items for the factor recruitment efficiency were adapted from [11,23]; and anthropomorphism was measured with scale items adapted from [8]. These scale items were enumerated on a Likert-type scale from 1 to 7 (1 strongly disagree to 7 strongly agree). The scale instruments are shown in Table 1, which demonstrates their satisfactory construct reliability and convergent validity.

Table 1. Measurement model.

Item	Loading	α	AVE	CR
AAI1: I predict that I will use artificial intelligence in recruitment.	0.871	0.873	0.798	0.922
AAI2: I anticipate that I will manage recruitment using artificial intelligence.	0.904			
AAI3: I highly intend to adopt artificial-intelligence-driven recruitment.	0.904			

Table 1. *Cont.*

Item	Loading	α	AVE	CR
ACC1: An artificial-intelligence-driven recruitment process is error-free.	0.831	0.928	0.824	0.949
ACC2: Use of artificial intelligence in the recruitment process improves its quality.	0.959			
ACC3: An artificial-intelligence-driven recruitment process reduces biases in the hiring process.	0.941			
ACC4: Artificial-intelligence-driven recruitment enhances the HR team's preciseness and accuracy.	0.895			
AUT1: Artificial-intelligence-driven recruitment is time-saving.	0.792	0.809	0.636	0.875
AUT2: Use of artificial intelligence in the recruitment process reduces the workload.	0.786			
AUT3: Use of artificial intelligence in the recruitment process is cost-effective.	0.828			
AUT4: Artificial-intelligence-driven recruitment assists HR professionals in making the right decisions.	0.783			
INT1: Artificial-intelligence-enabled recruitment can find the right candidate.	0.989	0.976	0.955	0.984
INT2: An artificial-intelligence-enabled recruitment process is sufficiently intelligent to screen potential candidates.	0.967			
INT3: I feel that an artificial-intelligence-driven recruitment process is competent in meeting HRs goals.	0.975			
PAN1: An artificial-intelligence-driven recruitment application is animated.	0.807	0.781	0.696	0.873
PAN2: An artificial-intelligence-driven recruitment application has its own mind.	0.821			
PAN3: An artificial-intelligence-driven recruitment application can make sense of emotions.	0.874			
PNT1: An artificial-intelligence-driven recruitment application provides real-time feedback.	0.923	0.909	0.786	0.936
PNT2: An artificial-intelligence-driven recruitment application is problem-solving.	0.864			
PNT3: An artificial-intelligence-driven recruitment application provides comparative solutions to my issues.	0.876			
PNT4: An artificial-intelligence-driven recruitment application can engage the right candidate for a competitive position.	0.883			
PSZ1: I can use an artificial-intelligence-enabled recruitment application as per my job requirements.	0.876	0.900	0.834	0.938
PSZ2: Using an artificial-intelligence-enabled recruitment application makes me feel like I am special.	0.938			
PSZ3: An artificial-intelligence-enabled recruitment application provides suitable solutions to my problems.	0.925			
REE1: An artificial-intelligence-enabled recruitment application meets organizational expectations.	0.950	0.903	0.838	0.939
REE2: Use of artificial intelligence in recruitment aids employees in completing tasks on time.	0.894			
REE3: The use of artificial intelligence speeds up the recruitment process.	0.900			
TRE1: An artificial-intelligence-enabled recruitment application can solve complex problems.	0.848	0.814	0.640	0.877
TRE2: An artificial-intelligence-enabled recruitment application assists HR professionals in the decision-making process.	0.757			
TRE3: Artificial-intelligence-driven recruitment applications can analyze large datasets.	0.800			
TRE4: An artificial-intelligence-enabled recruitment application offers real-time solutions.	0.793			

3.2. Sample Size and Data Collection

The research framework was tested with empirical responses retrieved from HR practitioners. Before data collection, the necessary sample size for this study was computed with a priori power analysis. Following the guidelines from [21], a medium effect size was selected with eight predictors in the sample power analysis, which revealed that 160 re-

sponses were required to test the research framework. A purposive sampling approach was selected in this study [21] to target HR practitioners, who held the knowledge we sought about AI-driven recruitment. Data were collected physically using survey questionnaires. Overall, 359 questionnaires were distributed among HR practitioners, who were requested to fill out the survey according to their understanding of AI-driven recruitment. Reminders and follow-up phone calls were used as tools to prompt the practitioners to complete this survey. Ultimately, we retrieved 336 valid empirical responses.

Participation in the AI recruitment survey was voluntary, and respondents were assured that their personal identity would not be disclosed. In our cross-sectional research design, data were collected at one point in time. After data collection, the respondents' profiles were examined with descriptive analysis, and factor reliability was tested through structural equation modeling. Descriptive analysis revealed that 33.6% of respondents were aged between 21 and 30 years, 41.1% between 31 and 40 years, and 25.3% between 41 and 50 years. Moreover, 14.3% of respondents had 1–5 years of experience in an HR department, 56% had 6–10 years of experience, and 29.8% had 11–15 years of experience, during which time they had assumed a senior recruitment officer role in the HR department. Among these HR professionals, 87.2% were male and 12.8% female.

4. Data Analysis

4.1. Common Method Bias

The common method bias issue is likely to occur in a survey-based study, and it is also likely to arise in a cross-sectional study where data are collected at once for exogenous and endogenous factors. Statistical and procedural interventions may address this issue. For instance, ref. [24] recommends that before data collection, the question order in the survey should be randomized to increase respondents' attention when answering the questions. Harman's single-factor analysis may also be employed [21,25]; this has a 40% threshold value, and the first factor's variance must be less than that threshold value for it to be concluded that the data are free of common method bias issues [21]. In this study, when the data were analyzed, the maximum variance explained by the first factor was 21%, which was substantially less than the threshold value. Therefore, we established that the data were free of common method bias.

4.2. Structural Equation Modeling

The data were analyzed with a modern statistical approach, structural equation modeling, which tests data in two stages. First, the convergent validity and reliability of factors are established, followed by their discriminant validity. Second, the relationship between hypotheses is tested through structural assessment. Indicator reliability is measured with the threshold value of 0.60 [20], factor reliability is achieved through Cronbach's alpha, and composite reliability follows the threshold value of 0.70 [20,26]. Similarly, the convergent validity of factors is determined based on the criterion that average variance extracted values should be greater than 0.50 [20]. Our results, illustrated in Table 1, showed satisfactory indicator reliability, factor reliability, and convergent validity.

Factors' discriminant validity, which ensures they measure unique and distinct concepts [25], is established with Fornell and Larcker's method [20,27] of following the pattern of the average variance extracted square root. The results in our case, as exhibited in Table 2, revealed satisfactory values for the average variance extracted, which ensured the factors' discriminant validity.

Discriminant validity was assessed with the cross-loading method, through which the data were cross-validated [28,29]. Following that, the indicator loadings were compared with the corresponding factor loadings [25]. The results revealed that none of the factor loadings

were greater than those of the other factors, which allowed us to establish that the factors were discriminant and measured unique concepts. The loading values can be seen in Table 3.

Table 2. Discriminant validity.

Factors	AAI	ACC	AUT	INT	PAN	PNT	PSZ	REE	TRE
AAI	0.893								
ACC	0.537	0.908							
AUT	0.488	0.236	0.798						
INT	0.839	0.447	0.422	0.977					
PAN	0.047	−0.049	−0.015	0.006	0.834				
PNT	0.898	0.463	0.415	0.796	0.107	0.887			
PSZ	0.612	0.204	0.307	0.460	0.028	0.500	0.913		
REE	0.523	0.295	0.333	0.375	0.110	0.497	0.306	0.915	
TRE	0.082	0.064	0.033	−0.011	0.007	0.061	−0.005	0.100	0.800

Table 3. Cross-loadings.

Factor	AAI	ACC	AUT	INT	PAN	PNT	PSZ	REE	TRE
AAI1	0.871	0.458	0.429	0.974	0.021	0.831	0.460	0.363	0.010
AAI2	0.904	0.545	0.438	0.630	0.057	0.775	0.551	0.566	0.123
AAI3	0.904	0.433	0.440	0.646	0.046	0.800	0.631	0.469	0.085
ACC1	0.491	0.831	0.233	0.535	−0.047	0.448	0.196	0.181	0.035
ACC2	0.523	0.959	0.232	0.374	−0.066	0.427	0.192	0.315	0.065
ACC3	0.507	0.941	0.203	0.397	−0.010	0.428	0.179	0.309	0.067
ACC4	0.413	0.895	0.184	0.305	−0.054	0.372	0.172	0.258	0.065
AUT1	0.387	0.216	0.792	0.353	0.001	0.340	0.221	0.354	−0.009
AUT2	0.380	0.210	0.786	0.358	0.005	0.341	0.206	0.289	0.010
AUT3	0.394	0.167	0.828	0.362	−0.055	0.326	0.274	0.253	0.032
AUT4	0.394	0.161	0.783	0.275	0.002	0.318	0.278	0.169	0.071
INT1	0.853	0.456	0.418	0.989	0.011	0.812	0.447	0.375	0.007
INT2	0.797	0.421	0.425	0.967	−0.017	0.755	0.440	0.370	-0.039
INT3	0.808	0.432	0.395	0.975	0.024	0.764	0.463	0.353	−0.002
PAN1	0.014	−0.048	−0.031	−0.016	0.807	0.087	−0.039	0.094	0.026
PAN2	0.082	−0.035	0.022	0.055	0.821	0.102	0.054	0.088	0.042
PAN3	0.023	−0.039	−0.026	−0.020	0.874	0.080	0.056	0.095	−0.047
PNT1	0.800	0.396	0.364	0.883	0.060	0.923	0.408	0.371	0.024
PNT2	0.826	0.475	0.389	0.562	0.131	0.864	0.465	0.567	0.113
PNT3	0.827	0.378	0.394	0.595	0.098	0.876	0.546	0.492	0.068
PNT4	0.720	0.390	0.318	0.798	0.089	0.883	0.338	0.312	0.002
PSZ1	0.581	0.183	0.291	0.475	0.035	0.465	0.876	0.313	−0.052
PSZ2	0.559	0.203	0.291	0.410	0.019	0.465	0.938	0.255	−0.010
PSZ3	0.534	0.172	0.258	0.371	0.021	0.437	0.925	0.267	0.051
REE1	0.455	0.214	0.317	0.340	0.098	0.441	0.290	0.950	0.073
REE2	0.540	0.375	0.319	0.371	0.080	0.511	0.283	0.894	0.103
REE3	0.424	0.197	0.273	0.309	0.130	0.397	0.266	0.900	0.096
TRE1	0.074	0.069	0.010	−0.022	0.079	0.046	−0.007	0.116	0.848
TRE2	0.048	0.094	0.059	−0.029	0.065	0.029	0.031	0.137	0.757
TRE3	0.064	0.061	0.020	0.003	−0.066	0.045	−0.037	0.059	0.800
TRE4	0.070	−0.006	0.028	0.006	−0.045	0.069	0.005	0.024	0.793

4.3. Hypothesis Analysis

Our hypotheses were examined using structural model assessment, which computes data with bootstrapping to mitigate data normality issues. Meanwhile, the bootstrapping process reveals t-statistics, path coefficients, and pathway significance. In our case, the results demonstrated that perceived interactivity positively impacted HR professionals' intention to adopt AI-driven recruitment (β = 0.494, SE 0.098, t-statistic 5.032, $p < 0.000$), confirming H1. Perceived intelligence was found to positively influence employees' intention to adopt AI-driven recruitment (β = 0.268, SE 0.077, t-statistic 3.491, $p < 0.000$), and hence H2 was confirmed. The relationship between personalization and employees' intention to adopt AI-driven recruitment was also confirmed (β = 0.192, SE 0.040, t-statistic 4.844, $p < 0.000$), upholding H3. Similarly, accuracy was revealed to positively impact employees' intention to adopt AI-driven recruitment (β = 0.128, SE 0.029, t-statistic 4.368, $p < 0.000$), and hence H4 was accepted. Automation was likewise shown to positively impact employees' intention to adopt AI-driven recruitment (β = 0.079, SE 0.022, t-statistic 3.607, $p < 0.000$), and hence H5 was confirmed.

The results then demonstrated that a real-time experience was positively related to employees' intention to adopt AI-driven recruitment (β = 0.045, SE 0.023, t-statistic 1.923, $p < 0.028$), and hence H6 was confirmed. Moreover, the relationship between employees' intention to adopt AI-driven recruitment and recruitment efficiency was confirmed (β = 0.518, SE 0.058, t-statistic 8.863, $p < 0.000$), and hence H7 was confirmed. These findings establish that the research framework is highly effective at measuring employees' intention to adopt AI-driven recruitment and recruitment efficiency. Altogether, the perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience explained substantial variance (R^2 0.897) in employees' intention to adopt AI-driven recruitment. Likewise, we found that recruitment efficiency is determined by employees' intention to adopt AI-driven recruitment and its perceived anthropomorphism, which explained a large part of the variance (R^2 0.293) in recruitment efficiency. These results affirmed that the research framework is statistically valid and capable of determining employees' intention to adopt AI-driven recruitment and its related efficiency. The statistical findings and hypothesis results are shown in Table 4.

Table 4. Hypothesis analysis.

Hypothesis	Path	β	STDEV	t-Statistic	Significance	Result
H1	PNT -> AAI	0.494	0.098	5.032	0.000	Supported
H2	INT -> AAI	0.268	0.077	3.491	0.000	Supported
H3	PSZ -> AAI	0.192	0.040	4.844	0.000	Supported
H4	ACC -> AAI	0.128	0.029	4.368	0.000	Supported
H5	AUT -> AAI	0.079	0.022	3.607	0.000	Supported
H6	TRE -> AAI	0.045	0.023	1.923	0.028	Supported
H7	AAI -> REE	0.518	0.058	8.863	0.000	Supported

4.4. Importance Performance Analysis

Although the results confirmed the efficacy of the outlined hypotheses in measuring the intention to adopt AI-driven recruitment, the importance of the path was yet to be assessed. To that end, importance performance analysis was performed, with recruitment efficiency as the outcome factor. Data were calculated with IPMA analysis, and the results revealed that the adoption of AI-driven recruitment was the most impactful factor in measuring recruitment efficiency. Therefore, it was inferred that the adoption of AI-driven

recruitment is essential to achieve efficiency in the recruitment process. Perceived interactivity was then found to be the second most important factor in determining recruitment efficiency. Beyond that, the importance of perceived intelligence was found to be considerable, followed by the personalization of AI recruitment applications. The perceived anthropomorphism, real-time experience, accuracy, and importance were shown to be the least important. Factors' importance levels are exhibited in Table 5 with the total effect and performance index values.

Table 5. Importance performance analysis.

Recruitment Efficiency		
Outlined Factors	**Total Effect**	**Performance Index**
Adoption of AI-enabled recruitment	0.518	61.879
Accuracy	0.066	61.655
Automation	0.041	71.303
Perceived intelligence	0.139	62.912
Perceived anthropomorphism	0.065	60.416
Perceived interactivity	0.256	60.716
Personalization	0.100	60.826
Real-time experience	0.023	77.713

4.5. Effect Size Analysis f^2

Effect size analysis was conducted in this study to understand the actual effects of the factors. This analysis reveals the effect of each factor in measuring the outcome factor. To assess effect sizes, researchers use three levels of effects, with threshold values of 0.02 for small, 0.15 for medium, and 0.35 for large effect sizes. Data were calculated with the PLS algorithm, and the results demonstrated that perceived interactivity had a substantial effect size in measuring the intention to adopt AI, while personalization and intelligence had medium effect sizes. For the second outcome factor, the results showed that perceived anthropomorphism had a small effect size, while the intention to adopt AI-driven recruitment had a large one, demonstrating that policymakers can boost recruitment efficiency through the adoption of AI-driven recruitment. Table 6 depicts the results for the effect sizes with the two outcome factors, namely, adoption of AI-enabled recruitment and recruitment efficiency.

Table 6. Assessment of the results of the effect size f^2 analysis.

Factors	**Results f^2**	**Size**
Adoption of AI-enabled recruitment		
Accuracy	0.120	Small
Automation	0.048	Small
Perceived intelligence	0.237	Medium
Perceived interactivity	0.761	Large
Personalization	0.259	Medium
Real-time experience	0.019	Small
Recruitment efficiency		
Adoption of AI-enabled recruitment	0.379	Large
Perceived anthropomorphism	0.006	Small

4.6. Moderating Analysis

Perceived anthropomorphism was conceptualized as a moderator construct in the relationship between the adoption of AI-driven recruitment and recruitment efficiency.

Moderating analysis was performed in two stages. First, the pathway significance was tested through a bootstrapping procedure. Second, the strength of the moderating relationship was measured with a simple slope analysis. Before data computation, the interaction effect was assessed using the product indicator approach. The data were bootstrapped for hypothesis testing. The results are depicted in Appendix A, revealing a significant moderation effect of perceived anthropomorphism between the adoption of AI-driven recruitment and recruitment efficiency (β = 0.118, SE 0.068, t-statistic 1.748, significant at 0.041). Therefore, H8 was accepted—that perceived anthropomorphism positively moderates the relationship between the adoption of AI-driven recruitment and recruitment efficiency. Furthermore, the simple slope graph revealed an ascending trend for perceived anthropomorphism PAN at +ISD when compared with other paths, including perceived anthropomorphism PAN at −ISD and perceived anthropomorphism PAN at +mean. As depicted in Figure 2, the simple slope graph results allowed us to establish that higher perceived anthropomorphism (PAN at +ISD) will strengthen the relationship between the adoption of AI-driven recruitment and recruitment efficiency.

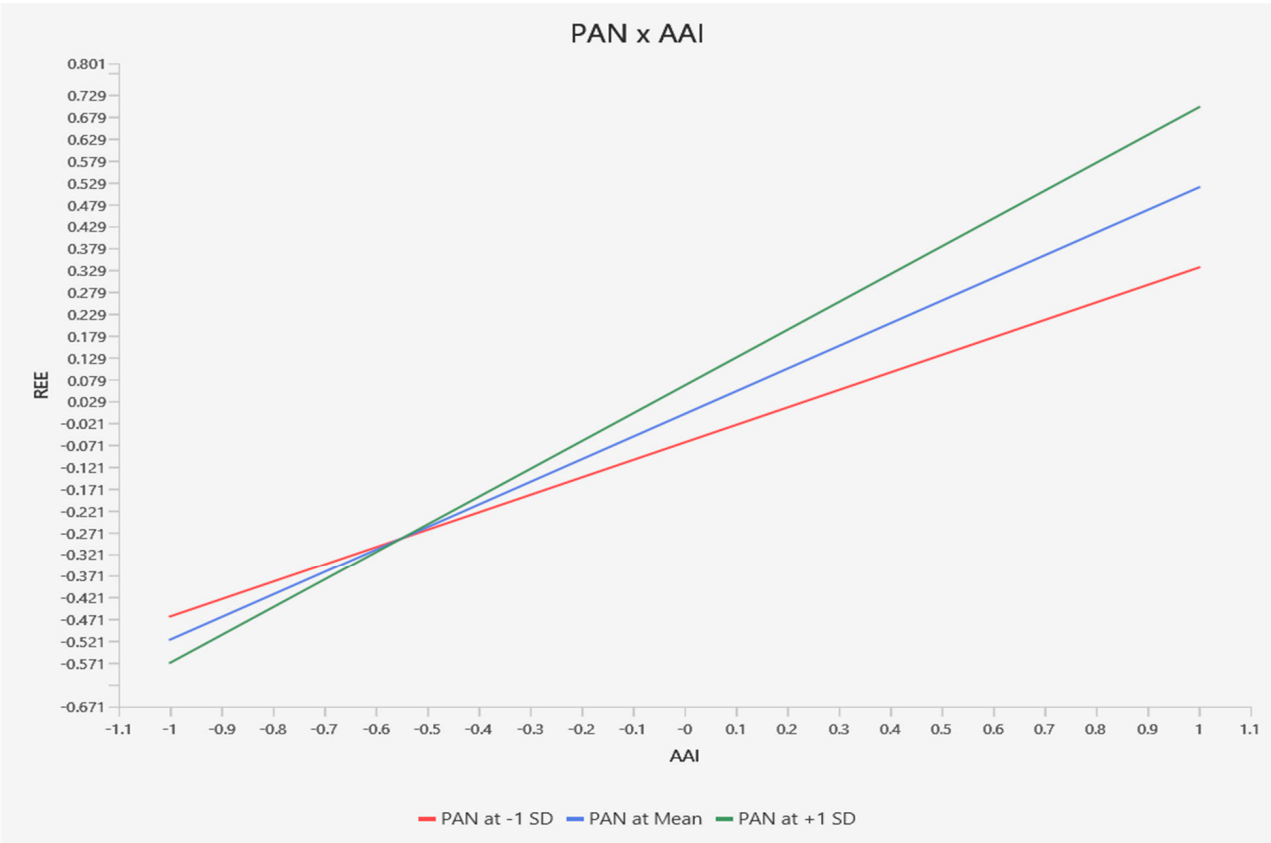


Figure 2. Simple slope grids.

5. Discussion

The Fourth Industrial Revolution is compelling organizations to adopt innovative AI-driven technologies in their business strategies. Among these strategies, recruitment is a big challenge, and so the current study examined the impact of artificial-intelligence-driven recruitment on recruitment efficiency. Prior studies have revealed that traditional brick-and-mortar recruitment policies are time-consuming, costly, and only somewhat effective [8,9]. Consequently, HR professionals are encouraged to adopt AI-enabled HR recruitment systems to improve the efficiency of the recruitment process [1,2]. The research framework of this study unveils key factors that influence employees' intention to adopt AI-

driven recruitment and thus save on complexity, time, and costs. Among those, perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience are the key factors that boost HR professionals' intention to adopt AI-driven recruitment. These factors were empirically tested with observations from HR professionals, and the results illustrated that perceived interactivity positively impacts their intention to adopt AI-driven recruitment, which is consistent with prior studies [6,7]. From this finding, we infer that AI-driven recruitment comprises characteristics of interpersonal attraction, enjoyment, and satisfaction, which raise employees' propensity to adopt AI-driven recruitment. Similarly, perceived intelligence is positively related to employees' intention to adopt AI-driven recruitment, in line with prior studies [8,9]. These findings demonstrate that AIs efficiency and autonomous characteristics assist employees in making the right hiring decisions and boost employees' confidence in selecting the right candidate.

The results were further compared with past studies, and we found consistency between personalization and employees' intention to adopt AI-driven recruitment [8,10,16]. Factors such as accuracy were also revealed to positively impact employees' intention to adopt AI-driven recruitment, in line with past studies [11,12,14,15]. Moreover, we found that automation positively impacts employees' intention to adopt AI-driven recruitment, which is consistent with the argument established by prior researchers [12,14]. Similarly, a real-time experience was shown to positively impact employees' intention to adopt AI-driven recruitment, in agreement with past studies [11,12,14,15]. Beyond that, this study established that perceived anthropomorphism positively moderates the relationship between the adoption of AI-driven recruitment and recruitment efficiency. In addition to that, the results also demonstrated that higher perceived anthropomorphism strengthens the relationship between the adoption of AI-driven recruitment and recruitment efficiency. In terms of the research framework's validity, this study confirmed that collectively the perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience explained substantial variance (R^2 0.897) in employees' intention to adopt AI-driven recruitment (i.e., substantially higher than in past studies). Finally, in regard to the importance of the factors, the IPMA results allow us to conclude that perceived interactivity, perceived intelligence, and personalization are the core factors enhancing HR professionals' intention to adopt AI-driven recruitment and hence improve the efficiency of the recruitment process.

5.1. Theoretical Contributions

This study contributes to the theory and the literature in several ways. For instance, it enriches the human resources literature by presenting a combined model with novel factors (perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience) and investigating HR professionals' intention to adopt an AI-driven recruitment system. Prior studies have rarely established an association between the outlined factors and the adoption of AI-driven recruitment. Filling that gap, the current study unveils novel factors that greatly influence employees' intention to adopt AI-driven recruitment and enhance efficiency in the recruitment process. The outcome factor—intention to adopt AI—is taken from the Theory of Planned Behavior; and our examination of employee behavioral intention in the context of AI-driven recruitment makes a large contribution to the information systems literature. Another unique contribution of this study lies in testing the moderating effect of perceived anthropomorphism between the adoption of AI-driven recruitment and recruitment efficiency. This study establishes that a higher level of perceived anthropomorphism will strengthen the relationship between the intention to adopt AI and recruitment efficiency and hence contribute to the anthropomorphism literature. Moreover, the theoretical validity of the research model is confirmed, with

substantial variance shown to be explained by exogenous factors. Chiefly, the perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience collectively explain 89.7% of the variance in employee behavioral intention to adopt AI-driven recruitment, which establishes the theoretical validity of the research model. These results underscore our assertion that adopting AI-driven HR practices may bring efficiency to organizations' recruitment processes.

5.2. Practical Contributions

This study signposts several directions in which HR professionals can improve the recruitment process using artificial intelligence. Starting from the research framework, the newly established research model has confirmed that factors such as perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience are central to determining employees' intention to adopt AI-driven recruitment. Broadly speaking, these findings indicate that AI-driven recruitment applications with perceived interactivity, perceived intelligence, personalization, accuracy, automation, and a real-time experience will raise employees' behavioral intention to adopt an AI-enabled recruitment system. More specifically, factors such as perceived interactivity, perceived intelligence, and personalization are considered greatly influential, given their large effect sizes, and hence can be expected to most significantly impact employees' intention to adopt AI-enabled recruitment applications. Furthermore, this research confirms that perceived anthropomorphism moderates the relationship between the adoption of AI-driven recruitment and recruitment efficiency. This provides support for the use of humanistic features for artificial intelligence forms such as robots and chatbots so they behave like agents and engage the right candidates for recruitment purposes. Such anthropomorphic characteristics of AI-driven recruitment applications are shown to have value in enhancing both the AI adoption intention among employees and the efficiency of the recruitment process.

6. Conclusions

Over the past two decades, artificial intelligence has become ubiquitous in organizations. Nevertheless, employees are hesitant to adopt AI-driven applications. Accordingly, this research investigated how artificial intelligence influences employees' intention to adopt AI-driven recruitment applications in organizations and bring efficiency to the recruitment process. To that end, this study developed a comprehensive research framework based on the literature. The research framework includes perceived interactivity, intelligence, personalization, accuracy, automation, real-time experience, and anthropomorphism and examines employees' intention to adopt an artificial-intelligence-driven recruitment system. To test this framework, data were analyzed using structural equation modeling. The empirical findings revealed that the perceived interactivity, perceived intelligence, personalization, accuracy, automation, and real-time experience collectively explained the substantial variance (R^2 89.7%) in employees' intention to adopt AI-driven recruitment. The effect size of each factor was assessed, and the results demonstrated a large effect of perceived interactivity on the intention to adopt AI, while personalization and intelligence had medium effects. These findings lead us to conclude that perceived interactivity, personalization, and perceived intelligence are the core factors that enhance employees' confidence to adopt AI-driven recruitment practices. Nevertheless, in the extended model, recruitment efficiency is determined by employees' intention to adopt AI-driven recruitment and perceived anthropomorphism. In the results, the intention to adopt AI-driven recruitment and perceived anthropomorphism explained R^2 29.3% of the variance in recruitment efficiency. To reduce the complexity of the model, IPMA analysis was conducted with recruitment efficiency as the outcome factor. The IPMA results revealed that perceived interactiv-

ity, perceived intelligence, personalization, accuracy, and automation are the core factors enhancing HR professionals' intention to adopt AI-driven recruitment and hence bring efficiency to the recruitment process. The moderating effect of anthropomorphism was also confirmed in the relationship between the adoption of AI-enabled recruitment and recruitment efficiency, with the results establishing that perceived anthropomorphism strengthens that relationship. This research uniquely provides a new framework for identifying opportunities to improve the recruitment process through artificial-intelligence-driven HR practices. Moreover, this research has unveiled that AI-driven recruitment improves the efficiency of the recruitment process, supporting HR professionals to hire the right people to achieve the organization's strategic goals.

Research Limitations and Future Directions

This study has certain limitations, which should be overcome in future research. First, this study is limited to the positive factors that impact individuals' intention to adopt AI and improve recruitment efficiency. Nevertheless, other factors negatively impact individuals' intention to adopt AI-enabled recruitment systems, such as technology anxiety, fear, security risks, and confidentiality while using AI-driven technology. Accordingly, future researchers should extend the current research framework to include suggested factors for more comprehensive results. Second, this study takes its outcome factor from the Theory of Planned Behavior, and for a different approach, we suggest that future researchers should include factors from the Task Technology Fit Model, such as task and technology characteristics. Third, this study is limited in its scope to the initial adoption of AI-driven recruitment among employees, while the post-adoption context is yet to be examined. In the future, extending the current research model to the ongoing intention setting will reveal factors that encourage employees to continue using AI-driven recruitment in the workplace. Finally, this study has a cross-sectional research design, which may limit the generalizability of the research model. In future studies, the research framework may be strengthened by testing it under a longitudinal design with different datasets.

Author Contributions: Conceptualization, A.F.D. and M.A.Y.; Methodology, A.F.D. and M.A.Y.; Software, A.F.D. and M.A.Y.; Validation, A.F.D. and M.A.Y.; Formal analysis, A.F.D. and M.A.Y.; Investigation, A.F.D. and M.A.Y.; Resources, A.F.D. and M.A.Y.; Data curation, A.F.D. and M.A.Y.; Writing—original draft, A.F.D. and M.A.Y.; Writing—review and editing, A.F.D. and M.A.Y.; Visualization, A.F.D. and M.A.Y.; Supervision, A.F.D. and M.A.Y.; Project administration, A.F.D. and M.A.Y.; Funding acquisition, A.F.D. All authors have read and agreed to the published version of the manuscript.

Funding: This work was funded by the University of Jeddah, Jeddah, Saudi Arabia, under grant No. (UJ-24-DR-20360-1). Therefore, the authors thank the University of Jeddah for its technical and financial support.

Institutional Review Board Statement: Not applicable.

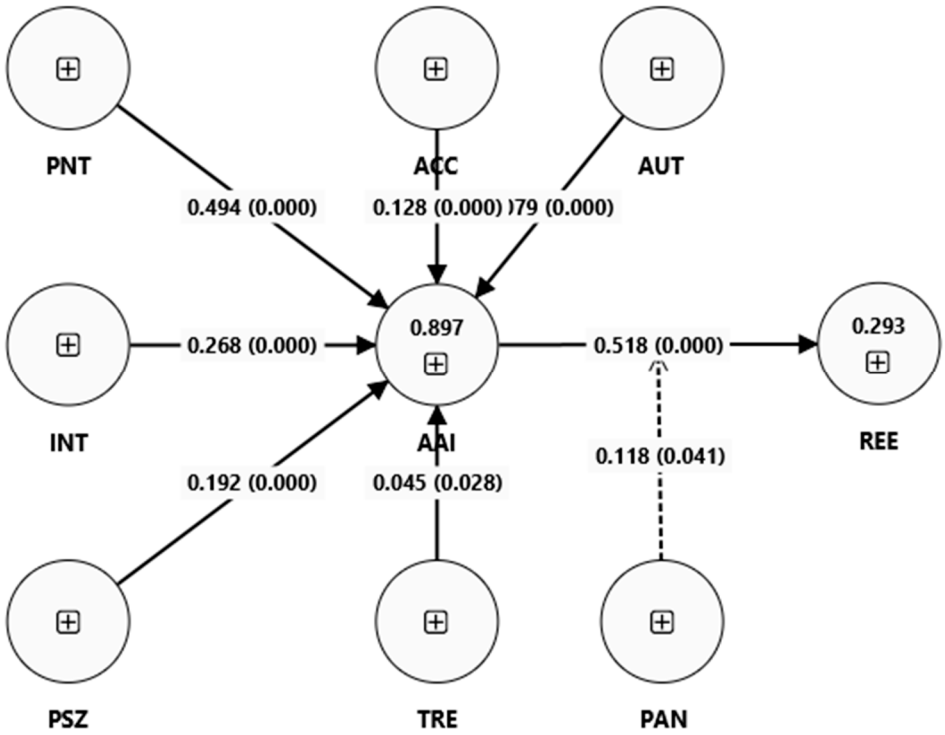
Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflict of interest.

Appendix A

Path Coefficient and Significance



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